



Brief Introduction



SENTINEL-X: TEAM STUDY & VIVA GUIDE

AI-Based Network Attack Forecasting from Network Traffic Data (SIH26153)

Department of Computer Science and Engineering • Buddha Institute of Technology, GIDA, Gorakhpur
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1. What is the Project? (The 60-Second Elevator Pitch)

The Real-World Analogy: Traditional Network Intrusion Detection Systems (like Snort or Suricata) behave like a sleeping security guard who only wakes up *after* a burglar has shattered the bank vault, grabbed the gold, and escaped. This is **reactive detection**.

Sentinel-X acts like an intelligent CCTV system watching the street outside. 30 seconds before any break-in happens, it notices a suspicious vehicle circling, masked individuals approaching the rear door, and power cables being cut. It alerts the security team: *'95% probability of an intrusion at the back door within 10 to 30 seconds.'* The doors are locked **before** the intruders even touch the handle. This is **predictive attack forecasting**.

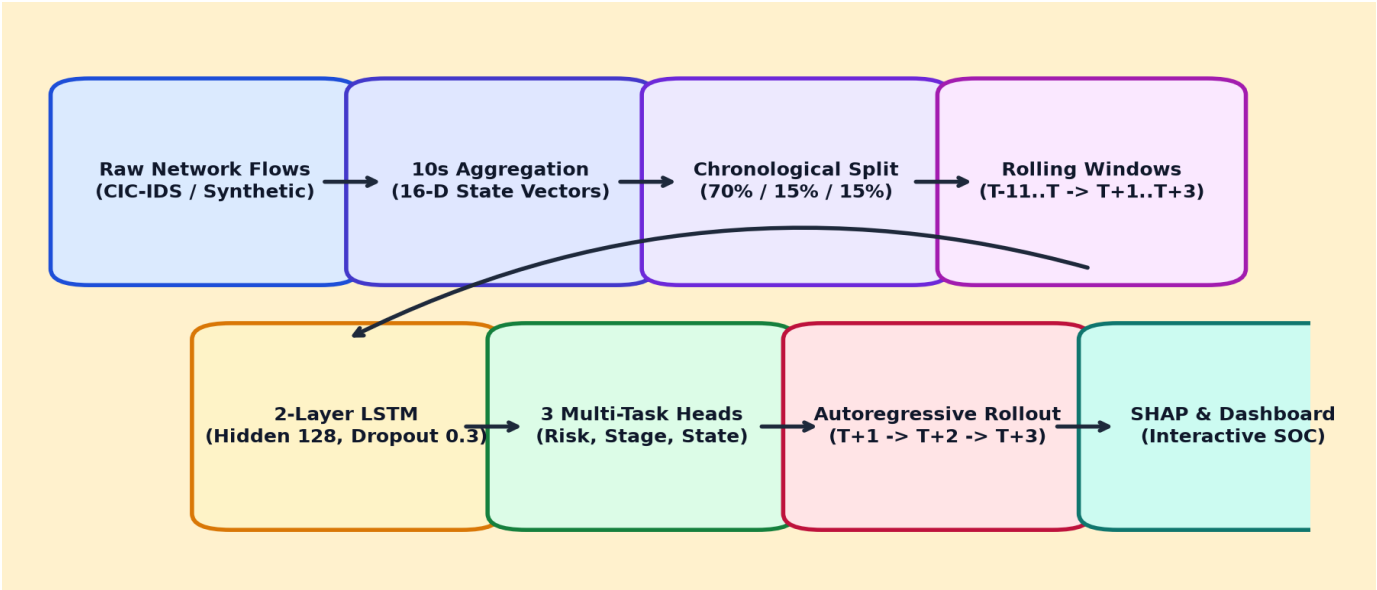
The Official Viva Definition:

"Sentinel-X is a defensive cybersecurity machine learning system addressing SIH problem statement SIH26153. Instead of inspecting isolated packets retrospectively, it aggregates continuous network flow telemetry into 10-second state vectors, and employs a Deep Recurrent LSTM World Model with 3 prediction heads to forecast: (1) multi-step attack risk probability at $T+1$, $T+2$, $T+3$; (2) MITRE ATT&CK; behavioral stages; and (3) continuous future network telemetry states, fully explained via SHAP XAI."

2. How Does Sentinel-X Work? (The 6-Step Pipeline)

Step 1: Telemetry Collection	Ingests network flows (IP addresses, ports, protocols, packet counts, bytes, TCP flags) from public defensive benchmarks (CIC-IDS-2017/2018) or deterministic simulated traffic.
Step 2: 10-Second Flow Aggregation	Packet-level inspection on 1 Gbps links causes buffer overflows. Sentinel-X aggregates flows into 10-second non-overlapping windows, deriving a continuous 16-dimensional telemetry state vector.
Step 3: Leak-Free Rolling Sequences	Extracts 12 historical time steps ($T-11 \dots T = 120$ seconds of history). Targets the next 3 steps: $T+1$ (10s), $T+2$ (20s), and $T+3$ (30s). Uses strict chronological 70/15/15 partitioning without shuffling to prevent temporal data leakage.

Step 4: 2-Layer LSTM World Model	A 2-layer LSTM backbone (128 hidden neurons, dropout 0.3) captures temporal sequence kinetics, projecting into 3 heads: Risk Head (BCE), Stage Head (Cross-Entropy), and State Head (MSE).
Step 5: Autoregressive Rollout Engine	At T+1, the model predicts next state s_{T+1} . It then feeds s_{T+1} back into the recurrent loop to forecast T+2, and s_{T+2} back to forecast T+3 without needing future ground truth.
Step 6: SHAP Explainable AI & Dashboard	Deploys SHAP (Shapley Additive exPlanations) to identify signed risk drivers (red) vs inhibitors (green). Visualized through an interactive 5-view Streamlit SOC dashboard.



3. The 16 Aggregated Network-State Features

Every team member must know these 16 features categorized into 4 groups:

#	Feature Name	Category	Cybersecurity Relevance / What it Detects
1	flow_count	Volumetric	Total flows in 10s. Sudden spikes reveal aggressive port scans or ping sweeps.
2	total_packets	Volumetric	Distinguishes high-volume floods from low-and-slow trickle beaconing.
3	packet_rate	Volumetric	Packets per second. Spikes during denial-of-service or brute force attacks.
4	total_bytes	Volumetric	Total volume transferred. Identifies large payload delivery or data exfiltration.
5	byte_rate	Volumetric	Bytes per second. Sustained elevated rates signify active outbound exfiltration.
6	duration_mean	Duration	Average flow duration. Very short in port scans; long in persistent sessions.
7	duration_std	Duration	Standard deviation of flow duration. Scripts/botnets exhibit rigid, low variance.
8	iat_mean	Timing	Mean Inter-Arrival Time. Low IAT = bursty scan; rigid periodic IAT = C2 beaconing.

9	fwd_pkt_len_mean	Dynamics	Forward packet size. Small during scans; large during exploit upload.
10	bwd_pkt_len_mean	Dynamics	Response packet size. Asymmetric fwd/bwd reveals auth failures/server rejects.
11	syn_ratio	Flag/Protocol	Ratio of SYN flags. Extreme values (> 0.5) indicate SYN floods or port probes.
12	rst_ratio	Flag/Protocol	Ratio of RST flags. Spikes when target hosts reject probes on closed ports.
13	ack_ratio	Flag/Protocol	Ratio of ACK flags. Normal benign traffic maintains high ACK ratios (> 0.85).
14	fin_ratio	Flag/Protocol	Ratio of FIN flags. Tracks graceful session termination frequencies.
15	tcp_ratio	Flag/Protocol	Proportion of TCP traffic in the 10-second aggregation window.
16	udp_ratio	Flag/Protocol	Proportion of UDP traffic (DNS amplification, streaming, UDP floods).

4. Conservative MITRE ATT&CK; Behavioral Taxonomy

Stage	Behavioral Label	MITRE Tactic	Telemetry Pattern
0	Benign Baseline	BENIGN	High ACK ratio (> 0.85), steady durations, expected HTTP/DNS traffic.
1	Reconnaissance	TA0043	Elevated SYN ratio (> 0.5), high flow counts, short flow durations.
2	Initial Access	TA0001	Repeated connection attempts to auth services (FTP/SSH/Patator), byte rate spikes.
3	Lateral Movement	TA0008	Internal subnet traversal across SMB (445), RDP (3389), WinRM (5985) ports.
4	Command & Control	TA0011	Periodic inter-arrival intervals (low variance) and consistent heartbeat payloads.

5. Verified Empirical Results (Memorize These Exact Numbers)

Evaluation Metric	Baseline (Logistic Reg)	LSTM World Model (T+1)	LSTM (T+2)	LSTM (T+3)
Risk Recall	97.8%	100.0% (Zero Misses)	100.0%	100.0%
Risk Precision	98.9%	96.7%	95.7%	94.6%
Risk F1-Score	0.9831	0.9834	0.9778	0.9721
False Positive Rate	20.0%	5.0% (75% Drop!)	8.0%	10.0%
Stage Accuracy	N/A (Binary Only)	62.8%	61.7%	60.6%

State Forecast MAE	N/A	0.8538	0.8645	0.8820
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Three Core Talking Points from the Results Table:

1. **100% Recall at T+1:** The LSTM World Model missed zero attacks on the immediate future horizon, compared to a 2.2% miss rate in the baseline.
2. **75% False Alarm Reduction:** The baseline had a 20% False Positive Rate (FPR); our model slashed it to 5%, directly preventing SOC alert fatigue.
3. **Predictive Degradation Curve:** F1 drops slightly from 0.9834 (T+1) to 0.9721 (T+3). *Examiners love this!* It proves the model is performing genuine closed-loop autoregressive rollouts where uncertainty naturally compounds.

6. Individual Team Member Roles & Viva Scripts

Member 1: Suraj Upadhyay (Roll No. 2405250100158) — Introduction, Problem Statement & System Architecture

Your Exact Presentation Script:

"Good morning respected guide Mr. Akash Gupta sir and respected examiners. I am Suraj Upadhyay. I present our project Sentinel-X, developed under SIH26153. Traditional NIDS are purely reactive—they alert after a system is compromised. Sentinel-X shifts security from reactive detection to predictive forecasting. We ingest 120-second rolling sequences of network telemetry and forecast attack risk 10, 20, and 30 seconds ahead. As shown in our architecture, raw network flows are cleaned, aggregated into 10-second windows, partitioned chronologically without leakage, and fed into our 2-layer LSTM World Model."

Expected Viva Question & Answer:

Q: Why is this defensive? A: Contains zero offensive payloads or scanners; operates purely on defensive incoming telemetry.

Member 2: Shruti Jaiswal (Roll No. 2405250100153) — Data Engineering, 16 Features & Chronological Windowing

Your Exact Presentation Script:

"Respected examiners, I am Shruti Jaiswal. On high-speed networks, packet-by-packet inspection causes buffer overflows. Sentinel-X solves this by aggregating flows into 10-second windows with 16 continuous statistical features: Volumetric, Duration, Packet Dynamics, and Protocol Flag Ratios. Crucially, we enforce strict chronological splitting (70% train, 15% val, 15% test) without shuffling to ensure zero future data leakage. We also map behaviors to a conservative 5-stage MITRE ATT&CK; taxonomy for triage."

Expected Viva Question & Answer:

Q: Why 10-second aggregation? A: Optimal sweet spot: avoids 1-second packet noise while preserving fast attack signature spikes.

Member 3: Suraj Prakash Chaudhary (Roll No. 2405250100156) — Deep Learning Architecture, 3 Heads & Multi-Task Loss

Your Exact Presentation Script:

"Respected examiners, I am Suraj Prakash Chaudhary. Our model is a 2-Layer LSTM World Model with 128 hidden units and 0.3 dropout. Unlike simple classifiers, a World Model learns how the environment evolves. It projects into three simultaneous heads: Risk Head (BCE), Stage Head (Cross-Entropy), and State Head (MSE). Our multi-task loss is: $L_{total} = 1.0 * L_{BCE} + 1.0 * L_{CE} + 0.2 * L_{MSE}$. Supervising the State Head provides auxiliary task regularization, forcing the network to learn genuine network dynamics. We perform closed-loop autoregressive rollouts."

Expected Viva Question & Answer:

Q: Why not a Transformer? A: Transformers have quadratic complexity $O(N^2)$ and need massive data. Our LSTM runs in < 20 ms on CPU with zero overfitting.

Member 4: Udai Kumar Srivastava (Roll No. 2405250100160) — Experimental Evaluation, SHAP XAI & SOC Dashboard Demo

Your Exact Presentation Script:

"Respected examiners, I am Udai Kumar Srivastava. Neural networks cannot be black boxes in a SOC. We integrate SHAP (Shapley Additive exPlanations) to show signed risk drivers (red bars) and inhibitors (green bars). In our benchmarks, our model achieved 100% recall at T+1 and cut false positives by 75% compared to the baseline. Our Streamlit SOC dashboard runs live and locally on our machine across 5 views: Executive Risk, Forecast Timeline, Threat Attribution, Model Performance, and System Diagnostics."

Expected Viva Question & Answer:

Q: Why does F1 taper from T+1 to T+3? A: Natural error compounding in autoregressive forecasting over extended horizons.

7. Top 10 Viva Questions & Bulletproof Answers

Q1: What is the main innovation of Sentinel-X?

Answer: It transitions network intrusion defense from reactive post-incident detection to proactive multi-step forecasting (T+1, T+2, T+3), buying critical mitigation time.

Q2: Why is it called a 'World Model'?

Answer: Because it predicts the future physical state of the network (s_{t+1}) and uses its own predicted state to roll out future predictions, simulating environment dynamics.

Q3: How do you prevent data leakage?

Answer: Strict chronological splitting (70% train, 15% val, 15% test) without shuffling. Scalers are fitted only on the training set. Sequence windows never overlap across split boundaries.

Q4: Why use an LSTM over XGBoost or Random Forest?

Answer: Intrusions are sequential processes (scanning -> access -> lateral pivot). Tree models treat time steps as independent rows and cannot perform autoregressive continuous state rollouts.

Q5: What is the purpose of the State Reconstructor Head?

Answer: It acts as an auxiliary task regularizer. Predicting the 16 physical features forces the LSTM latent space to understand network physics, mitigating overfitting on binary labels.

Q6: How does Sentinel-X reduce SOC alert fatigue?

Answer: It reduced the False Positive Rate from 20% (baseline) to 5% (a 75% reduction) and provides SHAP feature explanations so analysts don't waste time investigating false alarms.

Q7: What do the red and green bars mean in SHAP?

Answer: Red bars are Risk Drivers (features pushing risk UP, like high syn_ratio). Green bars are Risk Inhibitors (features anchoring risk DOWN, like normal ack_ratio).

Q8: How fast is inference?

Answer: Under 20 milliseconds on a standard laptop CPU, well within the 10-second operational telemetry window.

Q9: Is the model deployed on Vercel?

Answer: No. The PyTorch neural network runs locally (<http://localhost:8501>) because free serverless cloud platforms cannot run persistent PyTorch models. Vercel hosts the static documentation hub.

Q10: What are the primary future improvements?

Answer: Integration with automated SDN (Software-Defined Networking) OpenFlow controllers for instant firewall blocking, and ONNX quantization for microsecond edge router deployment.

8. The Exact Commands to Run the Project

1. **Activate Virtual Environment:** `.\.venv\Scripts\Activate.ps1`
2. **Run All 29 Unit/Integration Tests:** `pytest -v`
3. **Launch Interactive SOC Dashboard:** `streamlit run dashboard/app.py`
4. **Open Dashboard in Browser:** `http://localhost:8501`
5. **Live Documentation Hub:** `https://sentinel-x-lilac.vercel.app`